

JaeHyuk Son

Curriculum Vitae

Address:

Neuroengineering and Artificial Intelligence (NeuroAI) Laboratory,
Department of Electronics and Communications Engineering,
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RESEARCH INTERESTS

- **Core:** Computer Vision; Biomedical Signal Processing; Remote Photoplethysmography (rPPG); Contactless Blood Pressure Estimation
- **Methods:** Deep Learning; Transformer; Contrastive Learning; Representation Learning; Large Language Models (LLMs); Retrieval-Augmented Generation (RAG); Multi-Agent AI; Multimodal Learning
- **Applications:** Contactless Vital Sign Monitoring; Healthcare AI; Digital Health; Mental Healthcare AI; Edge AI

EDUCATION

M.S., Kwangwoon University, Seoul, Republic of Korea

Aug. 2024 – Aug. 2026 (Expected)

Department of Electronics and Communications Engineering

- Thesis: *A Physiology-Guided Lightweight Dual-Branch Model for Non-Contact Blood Pressure Estimation from Facial Videos*

B.S., Kwangwoon University

Mar. 2016 – Feb. 2023

Department of Electronics and Communications Engineering

EXPERIENCE

Research Experience:

Research Assistant, Neuroengineering and Artificial Intelligence Laboratory

Aug. 2024 – Present

Department of Electronics and Communications Engineering, Kwangwoon University

- Conducting research in computer vision, biomedical signal processing, contactless physiological measurement, and healthcare artificial intelligence.
- Participating in industry-academic collaborative research, AI healthcare product development, and international research projects.

RESEARCH PROJECTS

AI Champion 2026 – Domestic Track: Neuro-Sync Multi-Agent Mental Healthcare Handoff System

2026 – Present

Neuroengineering and Artificial Intelligence Laboratory, Kwangwoon University

- Developing a multi-agent AI platform for pre-clinical mental healthcare using Korean foundation large language models.
- Building RAG-based longitudinal memory from patient conversations and medical records.
- Developing structured clinician handoff reports and medical institution recommendation systems.
- Technologies: Multi-Agent AI, LangGraph, LangChain, RAG, FastAPI, Medical AI.

AI Champion 2026 – General Track: Neuro-Immerse Closed-Loop AI Agent for Alcohol Use Disorder

2026 – Present

Neuroengineering and Artificial Intelligence Laboratory, Kwangwoon University

- Developing a multimodal AI framework for detecting hidden alcohol craving.
- Integrating facial rPPG, voice biomarkers, wearable physiological signals, and LLM agents.
- Developing real-time personalized intervention through Edge AI.
- Technologies: Multimodal AI, LLM Agent, Edge AI, PyTorch, React Native.

AI Vital Sign Kiosk for Contactless Health Monitoring

2024 – 2026

Industry Collaboration with CUDO Communication

- Developed an AI-powered healthcare kiosk estimating heart rate, respiration rate, stress index, SpO₂, and blood pressure from facial videos.
- Designed real-time physiological signal estimation pipelines for commercial deployment.
- Successfully deployed the product and obtained GS (Good Software) Certification.

Mobile Application for Facial Video-Based Physiological Measurement

2025

Industry Collaboration with Inopia Tech

- Developed a mobile healthcare application for contactless physiological measurement using facial videos.
- Implemented AI inference pipelines optimized for mobile environments.

Arrhythmia Detection from Single-Lead ECG <i>Industry Collaboration with Inopia Tech</i> - Developed deep learning models for arrhythmia detection using single-lead ECG signals. - Designed preprocessing and classification pipelines for clinical ECG analysis.	2025
The 3rd Vision-Based Remote Physiological Signal Sensing (RePSS) Challenge <i>International Kaggle Competition</i> - Participated in the international RePSS Challenge on camera-based physiological signal sensing. - Contributed to the official RePSS Challenge Workshop paper. Achievement: 3rd Place.	2024
BERT-Based Fake News Detection and Chatbot Service <i>Goorm AI Bootcamp (Government-Funded Program)</i> - Developed a BERT-based fake news detection model and chatbot service. Achievement: Outstanding Project Award.	2023

SELECTED PUBLICATIONS

- J. H. Son** and Y.-S. Choi, “Non-Contact Blood Pressure Estimation from Face Videos via Physiology-Aware Contrastive Learning,” *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026. **Accepted.**
- S. H. Jho, D. H. Kim, **J. H. Son**, and Y.-S. Choi, “Physiology-Aware Multi-Branch Network for Fully Contactless Blood Pressure Estimation via Facial Remote Pulse Transit Time,” *International Joint Conference on Neural Networks (IJCNN)*, 2026.
- J. H. Son** and Y.-S. Choi, “Dual-Path Periodic Transformer for Enhancing Remote Photoplethysmography Estimation,” *International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC)*, 2025.
- J. H. Son** and Y.-S. Choi, “Transformer-Based Contactless Heart Rate Estimation Using Spatiotemporal Information from Facial Videos,” *Institute of Electronics and Information Engineers (IEIE) Summer Conference*, 2024.
- J. H. Son** and Y.-S. Choi, “Mamba-Based Physiological Feature Attention Framework for rPPG-Based Blood Pressure Estimation,” *Institute of Electronics and Information Engineers (IEIE) Summer Conference*, 2026.

RESEARCH PROFILE

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| Research Assistant, Neuroengineering and Artificial Intelligence Laboratory, Kwangwoon University. | 2024 – Present |
| Industry-Academic Collaborative Research with CUDO Communication and Inopia Tech. | 2024 – Present |

HONORS & AWARDS

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| Outstanding Poster Paper Award, The Korean Society of Medical and Biological Engineering (KOSOMBE). | 2024 |
| Outstanding Project Award, Goorm AI Bootcamp. | 2023 |
| Project Award, Violence and Non-Violence Situation Recognition using Korean Dialogues. | 2021 |

MILITARY SERVICE

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| Republic of Korea Army, Machine Gun Assistant Gunner, Honorably Discharged. |
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TECHNICAL SKILLS

- Programming Languages:**Python, C/C++, Java
- Deep Learning:** PyTorch, TensorFlow, CUDA, Transformer, Mamba, Contrastive Learning
- Computer Vision:** OpenCV, Remote Photoplethysmography (rPPG), Facial Video Analysis
- Biomedical AI:** ECG, PPG, VPG, APG, Blood Pressure Estimation, Physiological Signal Processing
- LLM & AI Systems:** LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Multi-Agent AI, FastAPI
- Platforms:** Linux, Docker, Git, AWS, PostgreSQL, React Native, Next.js